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A multi-state network to evaluate network reliability with maximal and minimal capacity vectors by using recursive sum of disjoint products

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ABSTRACT

This paper applies a multi-state network (MSN) model with multi-state capacity to evaluate network reliability with both maximal and minimal capacity vectors for real-life applications. To be specific, the proposed MSN model can be embedded in the practically related expert systems, especially in project networks, manufacturing, and supply chain. In the early works, the estimated network reliability with maximal and minimal capacity vectors and the exact value for the MSNs were measured based on an interval approach and inclusion and exclusion (IE) principle, respectively. Owing to the computational inefficiency, the early works are inappropriate to be integrated into practical expert systems. In this paper, a more efficient procedure based on the sum of disjoint products (SDP) principle calculates the exact network reliability. Additionally, a recursive function with simplified processes is developed to achieve the efficiency of reliability evaluation. By using a practical project network case with several numerical experiments, efficiency investigation reveals that the SDP-based procedure outperforms the IE-based approach in terms of efficiency under all the experiments. The proposed SDP-based procedure can achieve over 30 times (and reach up to 5,000 times) faster than the IE-based approach. That is, based on the proposed procedure, the reliability evaluation of practical applications such as project management can be implemented.

1. Introduction

Expert systems can emulate the decision-making ability of a human expert with proper models. In other words, modeling the real-life scenarios and problems into a certain model can assist the decision makers to identify the kernel of a problem and further make decisions. Therefore, an appropriate model is necessary to achieve the above goals for the real-life expert systems. The multi-state network (MSN) model is one of applicable models to represent and construct the real-world systems. The MSN model, which is problem-driven model, is a network consisting of components (edges and nodes) with multi-state capacities (i.e., multiple values). To reflect the real scenarios, including failures, maintenance requirements, and operational activities of the components in the MSN, each component in the MSN owns multi-state (multi-valued and stochastic) capacities. Each capacity owns a corresponding probability, which can be obtained from historical data. Then, the MSN model is a beneficial model to be embedded into practical expert systems for practical applications such as project management (Huang, Huang, & Lin, 2020), ERP system management (Chen & Lin, 2009), computer system evaluation (Yeh, 2019; Yeh and Fiondella, 2017), and supply chain management (Golpîra, 2020). For instance, a project can be characterized as an MSN in which every activity is presented as an arc with multiple values (including multi-valued duration and costs) as well as every node is utilized to separate the activity and its immediate predecessors. In practice, the expert systems with the MSN model are benefit to validate the possibility of the project management.

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To measure the performance of an MSN, previous studies (Bai et al., 2018; Chen & Lin, 2020; Niu et al., 2017; Yeh, 2017) evaluated network reliability for investigating the capability of MSNs. Network reliability is defined as the probability that an MSN can successfully transmit a required amount of demand under certain constraints (Bai et al., 2018; Yeh, 2017). Two approaches are widely applied to evaluate network reliability: lower-boundary-based method (Jane & Laih, 2017; Niu, Gao, & Sun, 2017) and upper-boundary-based method (Forghani-elahabad & Mahdavi-Amiri, 2016; Niu et al., 2017; Yeh, 2002). For the lower-boundary-based method, minimal capacity vectors, containing the minimal required capacity values for all the components, can satisfy demand are generated via minimal paths (Hernando et al., 2018). Network reliability is then derived by calculating the probability that capacity state equal to or larger than the minimal capacity vectors (Chen & Lin, 2020; Jane & Laih, 2017). Meanwhile, the upper-boundary-based method generates maximal capacity vectors whose demand cannot be met via minimal cuts (Golpîra, 2020). Note that each maximal capacity vector contains the maximal required capacity values for all the components. Then, network unreliability is evaluated which is the probability that the capacity state is equal to or less than maximal capacity vectors. Therefore, network reliability is 1 minus network unreliability (Forghani-elahabad & Mahdavi-Amiri, 2016; Niu et al., 2017; Yeh, 2002).

Some expert systems are modeled as the MSNs in the existing references (Chen & Lin, 2009; Yeh, 2019; Golpîra, 2020), and are devoted to solving such a problem with either the minimal or maximal capacity vectors (one-bounded capacity vectors). There are some methods that can be adopted for the calculation, including the inclusion–exclusion (IE) principle (Xue, 1985; Yarlagadda & Hershey, 1991), state-space decomposition (Aven, 1985), and approaches based on sum of disjoint products (SDP) (Bai et al., 2015; Datta & Goyal, 2017; Yeh, 2015). Bai et al. (2015) develops a recursive SDP algorithm with ordering heuristics, named RSDP approach, such that the probability can be calculated efficiently. Yeh (2015) also developed a recursive approach based on SDP by adding simplification procedures so as to reduce the number of multiplications and summations to increase the calculation efficiency. Datta and Goyal (2017) managed to minimize redundant computations in exact reliability computation based on SDP approach. According to previous works (Bai et al., 2015; Lin et al., 2017), the SDP-based approaches have been proven as efficiency for evaluating network reliability. Note that the aforementioned algorithms focus on using either minimal or maximal capacity vectors to calculate network reliability. Despite the well-studied methodologies for evaluating network reliability of an MSN, the previous works (Aven, 1985; Bai et al., 2015; Xue, 1985; Yarlagadda & Hershey, 1991) did not consider a situation in which the MSN contains both the multiple maximal and minimal capacity vectors simultaneously.

Note that, in some applications (especially, the management of the multi-state project network (Huang et al., 2019)), the maximal and minimal capacity vectors exist simultaneously under different conditions or criteria. To be specific, the time constraint and budget are commonly considered for the project management when evaluating reliability for a multi-state project network (Huang et al., 2019). The reliability is defined as the probability that a project can be completed under both time and budget constraints. The time constraint results in a set of maximal capacity vectors; while the budget results in a set of minimal capacity vectors. To assess the reliability of a multi-state project, both minimal and maximal flow vectors should be involved. Some scenarios that consider different constraints simultaneously (e.g., demand, time threshold, budget, and etc.) are also the cases of multiple maximal and minimal capacity vectors simultaneously. Therefore, the probability of all capacity vectors between the maximal and minimal capacity vectors must be studied by the proposed model. The study (Lin et al., 2017) have presented approximate solutions with interval estimation. That is, the exact network reliability is not evaluated in the work (Lin et al., 2017). Recently, a study (Huang et al., 2019) has proposed a breakdown approach to compute the exact network reliability with respect to the IE principle. However, when applying the IE principle, the computational time might increase rapidly when the studied networks become complicated or are large. Therefore, the efficiency of the exact network reliability calculation with the maximal and minimal capacity vectors should be improved.

To improve the computational efficiency, a novel procedure with the SDP principle is proposed in the novel MSN model to calculate the probability of all capacity vectors between the maximal and minimal capacity vectors. Because there are complex connections among multiple maximal and minimal capacity vectors, a completely feasible set is first decomposed to several simplified subsets. Network reliability is presented by these subsets and the SDP principle. A recursive function is further developed. In particular, the subsets can be narrowed down and further combined in the recursive function through simplified processes. The whole procedure is therefore proposed based on the above development to evaluate the exact network reliability. As mentioned above, the current SDP approaches (Bai et al., 2015; Datta & Goyal, 2017; Yeh, 2015) are unable to consider minimal and maximal flow vectors, simultaneously. To the best of our knowledge, the only approach to simultaneously consider minimal and maximal flow vectors for exact reliability evaluation is the IE-based algorithm in (Huang et al., 2019). Hence, to verify the outperformance of the proposed procedure against the IE-based approach (Huang et al., 2019), the computational efficiency is investigated.

The remainder of this paper is outlined as follows. The MSN model is constructed in Section 2 with nomenclatures and two crucial definitions. In addition, network reliability is formulated by decomposed subsets and the SDP principle. The recursive function is further developed. The simplified processes in the recursive function are developed in Section 3. Then, the whole procedure is created in Section 4. A simple MSN is shown to demonstrate the proposed procedure in Section 5. In Section 6, efficiency investigation is carried out by using a practical multi-state project network, which is a case study to validate the applicability to integrate the MSN model into an expert system. Finally, conclusions are depicted in Section 7.

2. MSN model and network reliability evaluation

This section intends to construct a multi-state network (MSN) model for the following network reliability evaluation with both maximal and minimal capacity vectors for real-life applications, such as project management. Let $\mathbf{T} \equiv (\mathbf{E}, \mathbf{M})$ be an MSN with r edges. To be specific, $\mathbf{E} = \{e_t \mid t = 1, 2, \ldots, r\}$ where e_t is the tth edges, and $\mathbf{M} = \{M_t \mid t = 1, 2, \ldots, r\}$ where M_t is the maximal capacity state of e_t. Each e_t presents several capacity states in terms of a certain probability distribution, which can be derived by the historical data. The following assumptions are met under every MSN **T**.

I. The capacity state of every e_t is a random variable with a certain probability distribution that can be derived by the historical data.
II. The capacity states of every e_t are statistically independent. That is, the failure of an edge does not affect another one.

2.1. Nomenclatures and definitions

Nomenclatures are shown as follows.

$X \leq Y$	$(x_1, x_2, \ldots, x_r) \leq (y_1, y_2, \ldots, y_r)$: $x_t \leq y_t$ for each $t = 1, 2, \ldots, r$.
$X < Y$	$(x_1, x_2, \ldots, x_r) < (y_1, y_2, \ldots, y_r)$: $X \leq Y$ and $x_t < y_t$ for at least one t.
$X \leq Y$	$(x_1, x_2, \ldots, x_r) \leq (y_1, y_2, \ldots, y_r)$: neither $X \geq Y$ nor $X < Y$.

A (current) capacity vector is termed $X = (x_1, x_2, \ldots, x_r)$ where x_t denotes the current capacity of e_t. Then, n minimal capacity vectors and m maximal capacity vectors, denoted $\underline{X}_i$ and $\overline{X}_j$ for $i = 1, 2, \ldots, n$ and $j = 1,$

2, …, m, contain all feasible X. That is, a set $\Omega = \{X \mid \underline{X}_i \le X \le \overline{X}_j, \forall i, j\}$ contains all the feasible X bound with $\underline{X}_i$ and $\overline{X}_j$. Definitions of both $\underline{X}_i$ and $\overline{X}_j$ are presented as follows.

Definition 1. *An X is an $\underline{X}_i$ if $X \in \Omega$ and $Y \notin \Omega$ with $Y < X$.*

Definition 2. *An X is an $\overline{X}_j$ if $X \in \Omega$ and $Y \notin \Omega$ with $Y > X$.*

2.2. Network reliability and developed recursive function

For an MSN **T**, network reliability R is defined as the probability of X satisfying certain constraints (i.e., the feasible X). It can be formulated as

$$R = Pr\{X|X \in \Omega\}. \tag{1}$$

In terms of $\underline{X}_i$ and $\overline{X}_j$, network reliability R is reformulated as

$$R = Pr\{X|X \in \Omega\} = Pr\{X|\leqslant X \leqslant, \forall i, j\}$$

There are several pairs of $\underline{X}_i$ and to contain all feasible X. The pairs may intersect such that complicated structures and connections exist between the multiple $\underline{X}_i$ and $\overline{X}_j$. For instance, with three maximal and minimal capacity vectors in Fig. 1, it can be found that the relationships of the pairs are intersected. Therefore, directly computing network reliability by listing all possible X among all $\underline{X}_i$ and $\overline{X}_j$ is time consuming (Huang et al., 2019).

To evaluate network reliability systematically, several subsets can be decomposed from Ω. Determined by $\overline{X}_j$, let $\mathbf{S}_j \equiv \bigcup_{i=1}^{n}\{X|\underline{X}_i \leqslant X \leqslant \overline{X}_j\}$ be a subset that is decomposed from Ω. Every $\mathbf{S}_j$ consists of the feasible X bound with one maximal capacity vector $\overline{X}_j$ and all minimal capacity vectors $\underline{X}_i$. Hence, network reliability R can be rewritten via $\mathbf{S}_j$ as follows:

$$R = Pr\{X|\underline{X}_i \leqslant X \leqslant \overline{X}_j, \forall i, j\} = Pr(\bigcup_{j=1}^{m} S_j) \tag{3}$$

According to Eq. (3), R is further presented as the form of the sum of disjoint products (SDP) principle:

$$R = Pr(\bigcup_{j=1}^{m} S_j) = \Pr\{X \in \mathbf{S}_1\} + \Pr(\{X \notin \mathbf{S}_1\}\{X \in \mathbf{S}_2\}) + \Pr(\{X \notin \mathbf{S}_1\}\{X \notin \mathbf{S}_2\}\{X \in \mathbf{S}_3\}) + \ldots + \Pr(\{X \notin \mathbf{S}_1\}\{X \notin \mathbf{S}_2\}\ldots\{X \in \mathbf{S}_m\}). \tag{4}$$

Initially, the first term $\Pr\{X \in \mathbf{S}_1\}$ with only one maximal capacity vector (i.e., $\overline{X}_1$) can be handily computed by utilizing existing approaches, including the RSDP approach with ordering heuristics (Bai et al., 2015), the IE principle (Hao, Yeh, Wang, Wang, & Sun, 2019; Xue, 1985; Yarlagadda & Hershey, 1991), and state space decomposition (Aven, 1985). Especially, the RSDP approach with ordering heuristics (Bai et al., 2015) is applied in this study.

The second term is the probability of the intersection of two events, $\{X \notin \mathbf{S}_a\}$ and $\{X \in \mathbf{S}_b\}$ with $a \neq b$, where $\{X \notin \mathbf{S}_a\}$ is the complement of event $\{X \in \mathbf{S}_a\}$. Thus, the second term can be rewritten as

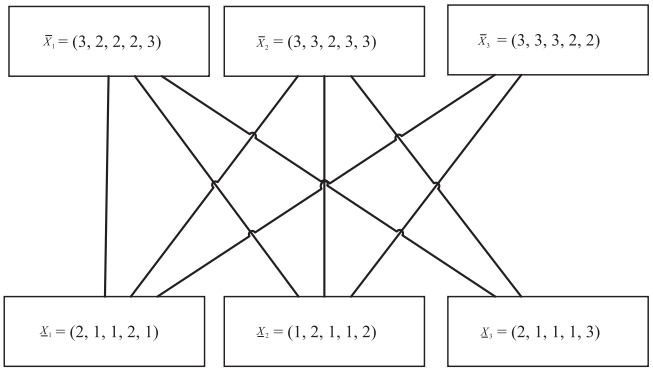


Fig. 1. An example of maximal and minimal capacity vectors.

$$\Pr(\{X \notin \mathbf{S}_a\}\{X \in \mathbf{S}_b\}) = \Pr(\{X \in \mathbf{S}_b\}) - \Pr(\{X \in \mathbf{S}_a\}\{X \in \mathbf{S}_b\}). \tag{5}$$

A special operator '⊗' for the capacity vectors is introduced. It is applied in the $\{X \in \mathbf{S}_a\}\{X \in \mathbf{S}_b\}$ with $a \neq b$ to generate a joint-upper capacity vector $\overline{Y}^{a,b}$ for simplifying the reliability calculation. The definition of operation '⊗' is provided as follows.

Definition 3. *Given two vectors X^1 and X^2, $X^1 \otimes X^2 \equiv \min_{v=1,2}(x_t^v)$ for all t.*

For instance, $X^1 = (2, 3, 1)$ and $X^2 = (1, 2, 3)$; then $X^1 \otimes X^2 = (1, 2, 1)$. The joint-upper capacity vector $\overline{Y}^{a,b}$ and a subset $\mathbf{S}_{a,b}$ are formulated, as follows.

$$\overline{Y}^{a,b} = \overline{X}_a \otimes \overline{X}_b \tag{6}$$

$$S_{a,b} = \bigcup_{i=1}^{n}\{X|\underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\} \tag{7}$$

Note that the indices, a and b, indicate that $\mathbf{S}_a$ and $\mathbf{S}_b$ are utilized in this case.

According to $\mathbf{S}_{a,b}$, $\Pr(\{X \notin \mathbf{S}_a\}\{X \in \mathbf{S}_b\})$ can be rewritten as

$$\Pr(\{X \notin \mathbf{S}_a\}\{X \in \mathbf{S}_b\}) = \Pr(\{X \in \mathbf{S}_b\}) - \Pr(\{X \in \mathbf{S}_a\}\{X \in \mathbf{S}_b\})$$

$$= \Pr(\{X \in \mathbf{S}_b\}) - \Pr(\{X \in \mathbf{S}_{a,b}\}). \tag{8}$$

Overall, $\Pr(\{X \in \mathbf{S}_{a,b}\})$ in Eq. (8) can be easily calculated by using the RSDP approach (Bai et al., 2015) because the subset $\mathbf{S}_{a,b}$ only contains a maximal capacity vector (i.e., $\overline{Y}^{a,b}$), instead of multiple $\overline{X}_j$.

The third term in Eq. (4) is the probability of the intersection of three events, $\{X \notin \mathbf{S}_a\}$, $\{X \notin \mathbf{S}_b\}$, and $\{X \in \mathbf{S}_c\}$ with $a \neq b \neq c$. In the same manner as Eq. (8), $\Pr(\{X \notin \mathbf{S}_a\}\{X \notin \mathbf{S}_b\}\{X \in \mathbf{S}_c\})$ can be rewritten as

$$\Pr(\{X \notin \mathbf{S}_a\}\{X \notin \mathbf{S}_b\}\{X \in \mathbf{S}_c\}) = \Pr(\{X \in \mathbf{S}_c\}) - \Pr(\{X \in \mathbf{S}_{a,c}\} \cup \{X \in \mathbf{S}_{b,c}\}). \tag{9}$$

Evidently, the third term with three events should consider the union of two events $(\{X \in \mathbf{S}_{a,c}\} \cup \{X \in \mathbf{S}_{b,c}\})$. Such a union can be recursively solved according to the SDP principle as follows.

$$\Pr(\{X \in \mathbf{S}_{a,c}\} \cup \{X \in \mathbf{S}_{b,c}\}) = \Pr(\{X \in \mathbf{S}_{a,c}\}) + (\Pr(\{X \in \mathbf{S}_{b,c}\} - \{X \in \mathbf{S}_{(a,c),(b,c)}\})), \tag{10}$$

where $\mathbf{S}_{(a,c),(b,c)}$ is generated by utilizing $\mathbf{S}_{a,c}$ and $\mathbf{S}_{b,c}$.

For the general case with k subsets: $\mathbf{S}_1$, $\mathbf{S}_2$, …, $\mathbf{S}_k$, a recursive function $\mathrm{R}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k)$ is defined as

$$\mathrm{R}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k) = \Pr(\{X \notin \mathbf{S}_1\}\{X \notin \mathbf{S}_2\}\ldots\{X \in \mathbf{S}_k\}) = \Pr(\{X \in \mathbf{S}_k\}) - \Pr(\bigcup_{j=1}^{k-1} \mathrm{S}_{j,k}). \tag{11}$$

To be specific, the term $\Pr(\{X \in \mathbf{S}_k\})$ can be computed by the RSDP approach (Bai et al., 2015). The following term, $\Pr(\bigcup_{j=1}^{k-1} \mathrm{S}_{j,k})$, is solved recursively by $\mathrm{R}(\mathbf{S}_{1,k}, \mathbf{S}_{2,k}, \ldots, \mathbf{S}_{k\text{-}1,k})$ with k-1 or fewer subsets. Note that the case with the fewer subsets is discussed in Section 3. The boundary condition for the recursive function $\mathrm{R}(\cdot)$ is $k = 2$ such that

$$\mathrm{R}(\mathbf{S}_1, \mathbf{S}_2) = \Pr(\{X \in \mathbf{S}_1\}) + (\Pr(\{X \in \mathbf{S}_2\}) - \Pr(\{X \in \mathbf{S}_{1,2}\})). \tag{12}$$

That is, Eq. (4) can be rewritten according to the recursive function R (·) as follows.

$$R = \Pr(\{X \in \mathbf{S}_1\}) + \mathrm{R}(\mathbf{S}_1, \mathbf{S}_2) + \mathrm{R}(\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3) + \ldots + \mathrm{R}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_m). \tag{13}$$

3. Simplified processes for the subsets and recursive function

Focused on a specific subset $\mathbf{S}_j = \bigcup_{i=1}^{n}\{X|\underline{X}_i \leqslant X \leqslant \overline{X}_j\}$, the relationship between a certain $\overline{X}_j$ and all $\underline{X}_i$ can be characterized by the index θ_i^j as follows.

$$\theta_i^j = \begin{cases} 1, & \text{if } \underline{X}_i \leqslant \overline{X}_j, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

For instance, there are two $\underline{X}_i$ and two $\overline{X}_j$: $\underline{X}_1 = (2, 2, 2)$, $\underline{X}_2 = (1, 3, 1)$,

$\overline{X}_1 = (2, 3, 2)$, and $\overline{X}_2 = (1, 4, 3)$. Indices $\theta_1^2 = 0$ and $\theta_2^2 = 1$ because $\underline{X}_1 \leqslant \overline{X}_2$ and $\underline{X}_2 < \overline{X}_2$, respectively.

In terms of the index θ_i^j, the subset $\mathbf{S}_j$ can be simplified so as to improve the computational efficiency according to the following theorem.

Theorem 1. *If* $\theta_i^j = 0$, $\{X \mid \underline{X}_i \leq X \leq \overline{X}_j\} = \varnothing$ *such that*

$$S_j = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_j\} \bigcup_{i:\theta_i^j=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_j\} \text{ for } j = 1, 2, \ldots, m.$$

Proof. Assume that $X^1 = (x_1{}^1, x_2{}^1, \ldots, x_r{}^1)$ is an $\underline{X}_i$ and $X^2 = (x_1{}^2, x_2{}^2, \ldots, x_r{}^2)$ is an $\overline{X}_j$ where $X^1 \leq X^2$ such that $\theta_i^j = 0$. Since $X^1 \leq X^2$, at least one x_p such that $x_p{}^1 > x_p{}^2$ and at least one x_q such that $x_q{}^1 < x_q{}^2$ where $p \neq q$. It concludes that $\{X \mid X^1 \leq X \leq X^2\} = \varnothing$ so that $\mathbf{S}_j$ is able to be reduced as $\bigcup_{i:\theta_i^j=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_j\}$. □

Extending from Theorem 1, we suppose there is a subset $\mathbf{S}_{a,b} = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\}$ generated from $\mathbf{S}_a$ and $\mathbf{S}_b$. For any pair of indices θ_i^a and θ_i^b, the subset $\mathbf{S}_{a,b}$ can also be reduced according to the following theorem.

Theorem 2. *If* $\theta_i^a + \theta_i^b \neq 2$, *then* $\bigcup_{i:\theta_i^a+\theta_i^b\neq 2} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\} = \varnothing$ *such that*

$$S_{a,b} = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\} = \bigcup_{i:\theta_i^a=1,\theta_i^b=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\} \quad (16)$$

Proof. Suppose that there are two capacity vectors: $X^1 = (x_1{}^1, x_2{}^1, \ldots, x_r{}^1)$ and $X^2 = (x_1{}^2, x_2{}^2, \ldots, x_r{}^2)$ are $\underline{X}_i$ with $X^1 \leq \overline{Y}^{a,b}$ or $X^2 \leq \overline{Y}^{a,b}$ such that $\theta_i^a + \theta_i^b \neq 2$. It concludes that $\bigcup_{i:\theta_i^a+\theta_i^b\neq 2} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\} = \varnothing$ so that $\mathbf{S}_{a,b}$ can be reduced as $\bigcup_{i:\theta_i^a=1,\theta_i^b=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{a,b}\}$. □

For the recursive function R(·) with inputted subsets $\mathbf{S}_l$: $\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k$, it is known that every $\mathbf{S}_l$ is bound with a joint-upper capacity vector $\overline{Y}^l$. Whenever possible, some inputted subsets in R(·) are combined according to the following theorem.

Theorem 3. *Suppose that there are* $\mathbf{S}_a$ *with* $\overline{Y}^a$ *and* $\mathbf{S}_b$ *with* $\overline{Y}^b$ *in* R(·). *If* $\overline{Y}^a \geq \overline{Y}^b$, *then* $\mathbf{S}^b$ *can be removed and* $\mathbf{S}^a$ *is updated as*

$$\mathbf{S}_a = \bigcup_{i:\theta_i^a=1,\theta_i^b=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^a\}. \quad (17)$$

Proof. Suppose that there are $\mathbf{S}_a = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^a\}$ and $\mathbf{S}_b = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^b\}$ with $\overline{Y}^a \geq \overline{Y}^b$ in the function R(·). Under the condition that $\overline{Y}^a \geq \overline{Y}^b$, we have $\mathbf{S}_a \cup \mathbf{S}_b = \mathbf{S}_a$ so as to remove $\mathbf{S}_b$. According to Theorem 2, $\mathbf{S}_a = \bigcup_{i=1}^{n} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^a\}$ can be reduced as $\bigcup_{i:\theta_i^a=1,\theta_i^b=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^a\}$. □

In order to combine the subsets in R($\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k$) with ($\overline{Y}^1, \overline{Y}^2, \ldots, \overline{Y}^k$), the following comparison approach is utilized to remove the non-maximal subsets and updating the maximal ones.

***Comparison** approach **with inputted vectors:** $\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k$*

Set $\lambda = \{\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k\}$. //$\lambda$ is used for storing the maximal subsets.
FOR $j_1 = 1$ to k and $j_1 \notin$ I
FOR $j_2 = j_1 + 1$ to k and $j_2 \notin$ I
IF $\overline{Y}^{j_1} \leqslant \overline{Y}^{j_2}$
$\mathbf{S}_{j_2} = \bigcup_{i:\theta_i^{j_1}=1,\theta_i^{j_2}=1} i : \theta_i^{j_1} = 1, \theta_i^{j_2} = 1$ //update $\mathbf{S}_{j_2}$.
$\lambda \leftarrow \lambda$ / $\{\mathbf{S}_{j_1}\}$ //remove $\mathbf{S}_{j_1}$.
BREAK //Next j_1
ELSE IF $\overline{Y}^{j_1} > \overline{Y}^{j_2}$
$\mathbf{S}_{j_1} = \bigcup_{i:\theta_i^{j_1}=1,\theta_i^{j_2}=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{j_1}\}$ //update $\mathbf{S}_{j_1}$.
$\lambda \leftarrow \lambda$ / $\{\mathbf{S}_{j_2}\}$ //remove $\mathbf{S}_{j_2}$.
END IF
END FOR
END FOR

4. Proposed procedure

Based on the developed recursive function with the simplified processes introduced in Section 3, the whole procedure is proposed to efficiently calculate network reliability as follows.

Input: $\mathbf{S}_j \equiv \bigcup_{i:\theta_i^j=1} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_j\}$ for $j = 1, 2, \ldots, m$.

Network reliability evaluation procedure

1. **RETURN [temp_rel] = recursive function** R($\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k$) with ($\overline{Y}^1, \overline{Y}^2, \ldots, \overline{Y}^k$)// Initially, ($\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_k$) = ($\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_m$) and ($\overline{Y}^1, \overline{Y}^2, \ldots, \overline{Y}^k$) = ($\overline{X}_1, \overline{X}_2, \ldots, \overline{X}_m$).
2. Calculate TM(1) = Pr{$X \in \mathbf{S}_1$} by using the RSDP approach (Bai et al., 2015). //TMs are temporary probabilities.
3. **FOR** $v = 2$ to k
4. For $a = 1, 2, \ldots, v$-1, generate the joint-upper capacity vectors $\overline{Y}^{a,v} = \overline{Y}^a \otimes \overline{Y}^v$.
5. Apply ***Comparison approach*** to obtain $\lambda = \{\mathbf{S}_{1,v}, \mathbf{S}_{2,v}, \ldots, \mathbf{S}_{v\text{-}1,v}\}$, which contains the maximal subsets.
6. **IF** $|\lambda| == 1$
7. TM(v) = Pr({$X \in \mathbf{S}_v$}) – Pr({$X \in \mathbf{S}_{1,v}$}) by using the RSDP approach (Bai et al., 2015).
8. **ELSE**
9. **RETURN [temp_rel] = recursive function** R($\mathbf{S}_{1,v}, \mathbf{S}_{2,v}, \ldots, \mathbf{S}_{v\text{-}1,v}$) with $\overline{Y}^{a,v}$ for $a = 1, 2, \ldots, v$-1. // Execute the recursive function again.
10. TM(v) = Pr({$X \in \mathbf{S}_{1,v}$}) – **temp_rel**
11. **END**
12. **END**
13. **RETURN temp_rel** = $\sum_{v=1}^{k}$ TM(v).

Output: R = **temp_rel**

Under each iteration with k subsets, the joint-upper capacity vectors are generated. We can recursively execute R(·) with the comparison approach to obtain network reliability R.

5. Illustrative example

An example is shown to demonstrate the proposed procedure. Capacity states of edges and the corresponding probability are provided in Table 1. Given that there are three minimal capacity vectors and three maximal capacity vectors: $\underline{X}_1 = (2, 1, 1, 2, 1)$, $\underline{X}_2 = (2, 1, 1, 1, 2)$, $\underline{X}_3 = (2, 1, 1, 1, 3)$, and $\overline{X}_1 = (3, 2, 2, 3, 3)$, $\overline{X}_2 = (3, 3, 2, 3, 3)$, $\overline{X}_3 = (3, 3, 3, 2, 2)$. The subsets $\mathbf{S}_1 = \bigcup_{i:1,2,3} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_1\}$, $\mathbf{S}_2 = \bigcup_{i:1,2,3} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_2\}$, and $\mathbf{S}_3 = \bigcup_{i:1,3} \{X | \underline{X}_i \leqslant X \leqslant \overline{X}_3\}$ are derived firstly. Network reliability R can be calculated according to the following steps.

Input: $\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3$

Network reliability evaluation procedure

1. Execute **R**($\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3$) with $\overline{X}_1, \overline{X}_2, \overline{X}_3$
2. Calculate TM(1) = Pr{$X \in \mathbf{S}_1$} = 0.4511 by using the RSDP approach.
4. For $\mathbf{S}_2$, generate $\mathbf{S}_{1,2} = \bigcup_{i:1,2,3} \{X | \underline{X}_i \leqslant X \leqslant \overline{Y}^{1,2}\}$ where $\overline{Y}^{1,2} = \overline{X}_1 \otimes \overline{X}_2 = (3, 2, 2, 3, 3) \otimes (3, 3, 2, 3, 3) = (3, 2, 2, 3, 3)$
5. Apply ***Comparison approach*** to obtain $\lambda = \{\mathbf{S}_{1,2}\}$.
7. Since $|\lambda| = 1$, TM(2) = Pr({$X \in \mathbf{S}_2$}) – Pr({$X \in \mathbf{S}_{1,2}$}) = 0.5638 – 0.4511 = 0.1128 by using the RSDP approach.
4.

(*continued on next page*)

Table 1
Capacity of the illustrative example.

Edge	state	Probability	Edge	state	Probability
e_1	3	0.70	e_4	3	0.20
	2	0.20		2	0.50
	1	0.10		1	0.30
e_2	3	0.20	e_5	3	0.25
	2	0.75		2	0.40
	1	0.05		1	0.35
e_3	3	0.30			
	2	0.65			
	1	0.05			

(*continued*)

Network reliability evaluation procedure

1. Execute $\mathbf{R}(\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3)$ with $\overline{X}_1$, $\overline{X}_2$,$\overline{X}_3$

For $\mathbf{S}_3$, generate $\mathbf{S}_{1,3} = \bigcup_{i:1,2,3}\{X|\underline{X}_i \leqslant X \leqslant \overline{Y}^{1,3}\}$ and $\mathbf{S}_{2,3} = \bigcup_{i:1,2}\{X|\underline{X}_i \leqslant X \leqslant \overline{Y}^{2,3}\}$ where $\overline{Y}^{1,3} = (3, 2, 2, 2, 2)$ and $\overline{Y}^{2,3} = (3, 3, 2, 2, 2)$.

5. Apply ***Comparison approach*** to obtain $\lambda = \{\mathbf{S}_{2,3}\}$ ($\overline{Y}^{1,3} < \overline{Y}^{2,3}$).

7. Since $|\lambda| == 1$, TM(3) = Pr($\{X \in \mathbf{S}_3\}$) – Pr($\{X \in \mathbf{S}_{2,3}\}$) = 0.4455 – 0.3118 = 0.1336 by using the RSDP approach.

13. **temp_rel** $= \sum_{v=1}^{3} \mathrm{TM}(v) = 0.4511 + 0.1128 + 0.1336 = 0.6975$.

Output: $R =$ **temp_rel** $= 0.6975$.

In the numerical example, network reliability $R = 0.6975$. The proposed procedure with the simplified processes evaluates five probability terms, including Pr($\{X \in \mathbf{S}_1\}$, Pr($\{X \in \mathbf{S}_2\}$), Pr($\{X \in \mathbf{S}_{1,2}\}$), Pr($\{X \in \mathbf{S}_3\}$), and Pr($\{X \in \mathbf{S}_{2,3}\}$). Then, the efficiency of the proposed procedure is further investigated and compared with the IE-based algorithm (Huang et al., 2019) in Section 6.

6. Effectiveness and efficiency investigation

Since the proposed procedures apply a certain recursive function, the theoretical time complexity is not easy to be presented, a practical project management case with both the maximal and minimal capacity vectors from (Huang et al., 2019) is provided in Section 6.1 to validate the accuracy and effectiveness of the proposed procedure. More importantly, this project network shows the applicability to integrate the proposed MSN model into an expert system. The multi-state project network is modeled as an MSN, and the efficiency of the proposed procedure is further investigated based on the same project management case in Section 6.2, compared with the IE-based approach in (Huang et al., 2019). Besides, more experiments are provided in Section 6.3 to examine the different scales of the MSNs by using the proposed algorithm.

6.1. Effectiveness investigation

The early work (Huang et al., 2019) studied a project of the construction of the new manufacturing lines in Chengdu, China. The project of the construction can be presented as an MSN with 13 main activities, shown in Fig. 2. The durations (days) and the corresponding cost (CNY) of each activity are displayed in Table 2.

Consider the same constraints in (Huang et al., 2019): the duration is 135 days and budget is 118,000. We can obtain 16 maximal and 9,490 minimal capacity vectors by the method proposed in (Lin et al., 2017). Then, network reliability R is 0.906411 by using the proposed procedure, which is the identical value in literature (Huang et al., 2019). Hence, the accuracy of the proposed procedure is proved.

Table 2
Durations, costs, and probabilities of each activity.

Arc	Activity	Duration (Days)	Cost (CNY)	Probability
a_1	Confirmation of the manufacturing lines	3	4,500	0.20
		5	3,000	0.40
		7	1,500	0.40
a_2	Confirmation and collection of the tools	2	8,400	0.10
		7	4,200	0.70
		10	3,000	0.20
a_3	Establishment of the labor requirement	7	4,500	0.10
		14	3,000	0.60
		21	1,500	0.30
a_4	Issue of a purchase order	14	2,400	0.10
		28	1,200	0.70
		42	600	0.20
a_5	Shipment of the tools	20	20,000	0.70
		30	15,000	0.30
a_6	Establishment of the SOP	7	1,800	0.20
		14	1,200	0.70
		18	600	0.10
a_7	Preparation of machines and materials	12	7,200	0.05
		14	6,000	0.90
		16	4,800	0.05
a_8	Custom declaration	14	1,200	0.20
		21	600	0.80
a_9	Shipment of machines and materials	21	40,000	0.70
		31	30,000	0.30
a_{10}	Construction of the first line in floor A	11	10,200	0.05
		18	9,000	0.05
		25	7,400	0.90
a_{11}	Construction of the second line in floor A	5	10,200	0.05
		12	9,000	0.05
		19	7,400	0.90
a_{12}	Construction of the first line in floor B	7	10,200	0.05
		14	9,000	0.05
		21	7,400	0.90
a_{13}	Construction of the second line in floor B	5	10,200	0.05
		12	9,000	0.05
		19	7,400	0.90

6.2. Efficiency investigation

The early work (Huang et al., 2019) proposed an algorithm based on the IE principle to evaluate network reliability with both maximal and minimal capacity vectors. The computational terms obtained by the IE principle might increase exponentially as the capacity vectors increase. In this section, we compare the efficiency of the proposed procedure, named the SDP-based procedure, with that of the IE-based procedure (Huang et al., 2019). The programs of both the algorithm were developed by MATLAB programming language and executed on a personal computer with Core™ I7-8700 3.2 GHz and 16-GB RAM.

In terms of the two algorithms' efficiency, we focus on the required

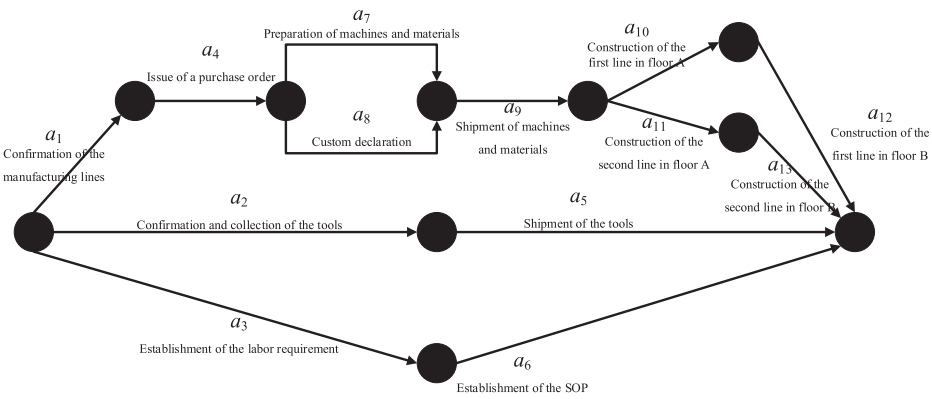


Fig. 2. The project of construction of the new manufacturing lines.

computational time for the different number of maximal and minimal capacity vectors. A hypothetical multistate network system with 20 components is considered. Each component has six capacity states, from zero to five; the state distributions of all the components are set to be equal. Specifically, the state distribution vector p is set to be

$$p = (0.1, 0.1, 0.1, 0.1, 0.1, 0.5). \quad (18)$$

The maximal and minimal capacity vectors are randomly generated to validate the applicability and scalability of the proposed procedure. For efficiency investigation in the experiments, there are 15, 20, 25, and 30 maximal as well as 15, 20, and 25 minimal capacity vectors applied to execute the SDP-based procedure and the IE-based procedure (Huang et al., 2019).

Every experimental scenario is executed five times, and the average CPU time is presented in Table 3. The CPU times of the SDP-based procedure and the IE-based procedure (Huang et al., 2019) are denoted as υ_1, and υ_2, respectively. From the experimental results, it is observed that the SDP-based procedure outperforms the IE-based procedure (Huang et al., 2019) in all cases according to the ratio $\lambda = \upsilon_2 / \upsilon_1$. Under$|\overline{X}_j| = 15$, $|\overline{X}_j|$=20, $|\overline{X}_j|$=25, the SDP-based procedure is approximately 30, 440, and 5,000 times faster than the IE-based procedure (Huang et al., 2019), respectively. Up to $|\overline{X}_j| = 30$, the IE-based procedure (Huang et al., 2019) is unable to complete the computation in 48 hrs. Thus, the CPU time of the IE-based procedure (Huang et al., 2019) increases rapidly as the problem becomes complicated. Conversely, the CPU time of the SDP-based procedure only increases steadily under similar conditions. Note that the computational time would increase when either the number of $\overline{X}_j$ or $\underline{X}_i$ increases. The above observations can be proved by viewing Fig. 3, which shows the average λ under the same number of the maximal capacity vectors. However, we can observe that the procedures are more sensitive to the number of the $\overline{X}_j$ than to $\underline{X}_i$. The computational time only increases slightly when the number of $\underline{X}_i$ increases by using the SDP-based procedure.

Table 3
Experimental results.

The number of arcs is 13		$\|\underline{X}_i\| = 15$	$\|\underline{X}_i\| = 20$	$\|\underline{X}_i\| = 25$	Average
$\|\overline{X}_j\| = 15$	CPU time of the SDP-based procedure (s)υ_1	0.198739	0.296443	0.404026	
	CPU time of the IE-based procedure (s)υ_2	6.73929	10.15239	14.52738	
	$\lambda = \upsilon_2 / \upsilon_1$	33.91031	34.24738	35.95658	34.705
$\|\overline{X}_j\| = 20$	CPU time of the SDP-based procedure (s)υ_1	0.493219	0.72548	1.02237	
	CPU time of the IE-based procedure (s)υ_2	216.698	325.1134	450.801	
	$\lambda = \upsilon_2 / \upsilon_1$	439.3546	448.1355	440.9373	442.809
$\|\overline{X}_j\| = 25$	CPU time of the SDP-based procedure (s)υ_1	1.408308	2.108217	2.877257	
	CPU time of the IE-based procedure (s)υ_2	7100.757	10468.58	14296.84	
	$\lambda = \upsilon_2 / \upsilon_1$	5042.05	4965.608	4968.912	4992.190
$\|\overline{X}_j\| = 30$	CPU time of the SDP-based procedure (s)υ_1	5.2610	6.35753	7.56853	
	CPU time of the IE-based procedure (s)υ_2	NA*	NA**	NA**	
	$\lambda = \upsilon_2 / \upsilon_1$	–	–	–	–

* Not complete in 48 hr.
** Out of memory in 24 hr.

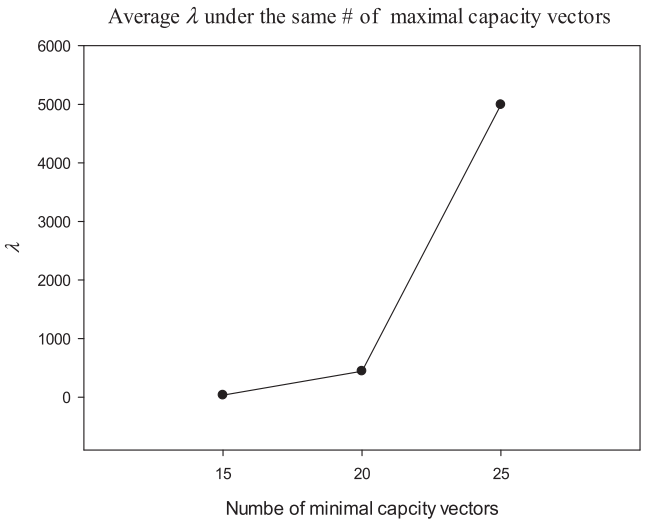


Fig. 3. The comparison char of average λ under the same number of $|\overline{X}_j|$.

6.3. More experiments

In order to learn the impact of the MSN scales for the proposed algorithm and the IE-based procedure (Huang et al., 2019), we randomly generate $\overline{X}_j$ with the different number of arcs under the randomly generated $\underline{X}_i$ where $|\underline{X}_i| = 20$. In detail, the number of $\overline{X}_j$ (resp. the arcs) is designed as 15, 20, 25 ((resp. 15, 20, 25). Hence, there are nine experimental combinations shown in Table 4. Note that even though some experimental scenarios are the same as the experiments in Table 3, the relationships of the capacity vectors are different such that the CPU times of the procedures are changed.

From the experimental results, the number of the arcs would also affect the efficiency of the procedures. When increasing to 25 arcs, the CPU time of the proposed procedure would increase significantly. However, the proposed procedure can obtain network reliability for all the experiments in reasonable time. For example, in the most complicated case, the CPU time of the proposed procedure is about 21.6174 s. On the other hand, the IE-based procedure cannot achieve the calculation after 24 h.

7. Conclusions

This paper intends to model real-life expert systems (such as

Table 4
More experiments.

$\|\underline{X}_i\| = 20$		$\|\overline{X}_j\| = 15$	$\|\overline{X}_j\| = 20$	$\|\overline{X}_j\| = 25$
# of arcs = 15	CPU time of the SDP-based procedure (s)υ_1	0.3246	1.5571	8.0936
	CPU time of the IE-based procedure (s)υ_2	0.8919	28.4112	887.6811
	$\lambda = \upsilon_2 / \upsilon_1$	2.7477	18.2462	109.6769
# of arcs = 20	CPU time of the SDP-based procedure (s)υ_1	0.3545	2.6381	10.4785
	CPU time of the IE-based procedure (s)υ_2	1.8477	27.5903	928.8891
	$\lambda = \upsilon_2 / \upsilon_1$	5.2121	10.4584	88.6471
# of arcs = 25	CPU time of the SDP-based procedure (s)υ_1	2.6835	5.2826	21.6174
	CPU time of the IE-based procedure (s)υ_2	NA*	NA*	NA*
	$\lambda = \upsilon_2 / \upsilon_1$	–	–	–

* Not complete in 24 hr.

manufacturing, supply chain, and project systems) as the multi-state networks (MSNs) and to evaluate network reliability with maximal and minimal capacity vectors for the decision-making scenario. Note that network reliability is defined as the probability to satisfy a certain demand. For instance, a multi-state project network considering time and budget constraints is modeled as an MSN. Its performance (i.e., network reliability) is evaluated for the project management by giving maximal capacity vectors to fulfill the budget constraint and minimal capacity vectors to fulfill the time constraints. In other words, network reliability for the multi-state project network is a performance indicator to comprehend the probability to satisfy the assigned time and budget constraints of a project.

In detail, the only existing approach (Huang et al., 2019) is based on the inclusion and exclusion (IE) principle to evaluate network reliability with the maximal and minimal capacity vectors that shows a severe disadvantage of inefficiency in computation time. To overcome the efficiency issue, this study proposes an MSN model to calculate network reliability with both maximal and minimal capacity vectors by applying the sum of disjoint products (SDP) principle with a developed recursive function. In particular, the simplified processes in the recursive function are proposed for computational efficiency. In the simplified processes, some subsets can be combined, and the number of the minimal capacity vectors is reduced significantly. A practical project management case is presented and experimented in Section 6 which is a case about the project management of the construction of the new manufacturing lines. The experimental results show that the proposed procedure achieves a much better efficiency than the IE-based procedure. The proposed SDP-based procedure can achieve over 30 times (and reach up to 5,000 times) faster than the IE-based procedure.

This proposed model has successfully presented a more efficient and analytic procedure with the recursive function to evaluate network reliability with both maximal and minimal capacity vectors. Several valuable issues are worth being investigated in the future. For example, in addition to the analytic approach, the simulation technique is widely applied to upper- and lower-boundary-based reliability evaluation because it performs better efficiency in very large-scale networks. Besides, the orthogonalization is a possible approach to generate new orthogonal vectors so as to simplify the proposed procedure. On the other hand, network reliability of the MSNs is calculated based on the minimal and maximal capacity vectors. Therefore, MSNs can be also presented by structure functions, which describes the connects of the component according to the related references (Lisnianski et al., 2010; Lisnianski et al., 2021; Natvig, 2010; Zaitseva and Levashenko, 2013). Hence, there might be a future research by calculating network reliability with the minimal and maximal capacity vectors by using the above functions.

CRediT authorship contribution statement

Ding-Hsiang Huang: Writing – original draft, Methodology. **Ping-Chen Chang:** Validation, Data curation. **Yi-Kuei Lin:** Writing – review & editing, Conceptualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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